Working Paper

From AI Diffusion to AI Capacity: A Framework, Diagnostic Tool, and Research Agenda for the Global South

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Abstract

The spread of AI across the Global South is frequently read as evidence of adoption, yet widespread use of AI tools reveals little about who controls the models, infrastructures, data pipelines, and contractual terms through which that use occurs. This paper argues that AI diffusion and AI capacity are analytically distinct, and that conflating them obscures the real terms on which countries participate in the AI economy. It develops a conceptual framework built on six primary entry points (markets, infrastructures, state systems, resource and industrial regimes, language ecologies, and development agendas) through which AI first becomes economically and institutionally consequential, and defines AI capacity across five interrelated components (computational, data and language, resource and infrastructural, institutional, and developmental) as the ability to access, build, adapt, govern, and sustain the conditions of AI development and use. Building on this framework, the paper offers a diagnostic tool: five recurring gaps: adoption without control, infrastructure without access, data without usability, skills without institutions, and pilots without systems, through which diffusion fails to convert into durable capacity, applied comparatively to India and South Africa as two distinct configurations of capacity formation. The paper closes with a six-question research agenda - entry point, capacity formation, connection, place and agency, bargaining power, and development alignment. The intention is to reorient AI governance research in the Global South from adoption and readiness rankings toward the political economy of capacity formation. The paper is structured to be used at three levels: as a conceptual vocabulary, as a standalone diagnostic tool applicable beyond the two cases discussed, and as an open research agenda - each usable independently of the others.

Keywords: artificial intelligence; AI capacity; AI governance; Global South; digital infrastructure; technological dependence; institutional capacity; digital sovereignty.

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1. Where does AI begin in a country? Entry points into the political economy of AI capacity

AI does not begin in a country at a single point. Long before it becomes the subject of a national strategy, it enters existing markets, infrastructures, state systems, resource regimes, language ecologies and development programmes. The route through which it first becomes consequential shapes the actors who gain influence, the institutions drawn into governance, and the early distribution of capacity, dependency and bargaining power.

Visible use is therefore an unreliable guide to a country's position in the AI economy. AI tools may spread while control over models, data systems, infrastructure and contractual terms remains elsewhere. Cloud regions, data centres or language resources may also develop before AI attracts sustained public attention. These trajectories represent different forms of incorporation, with different consequences for what countries can build, govern and negotiate.

The relevant question is where AI first acquires economic, institutional or infrastructural significance. In some countries, commercial demand leads; elsewhere, AI develops through public digital systems, infrastructure investment, industrial applications, language technologies or development programmes. Identifying the primary entry point reveals the terrain on which later governance choices are made.

Entry points overlap and evolve. Commercial demand can stimulate cloud investment; data-centre expansion brings energy, water and land-use decisions into AI governance; public-sector adoption raises questions of procurement, data management and administrative expertise. Language initiatives can open routes into education and public services, while development projects can consolidate research and policy networks. The initial route does not determine the entire trajectory, but it influences which actors and dependencies become established before a coordinated national approach emerges.

Six entry points recur across national and regional trajectories: markets, infrastructures, state systems, resource and industrial regimes, language ecologies and development agendas.

1.1. Entry through markets: AI as service, subscription and platform layer

Markets provide the fastest route into AI for many countries. Organizations purchase tools and services developed elsewhere and integrate them into existing operations. This route depends less on domestic model development than on connectivity, purchasing power, organizational demand and vendor relationships.

Market entry foregrounds adoption, but its political-economic significance lies in value capture and control. Widespread use may strengthen global cloud providers, software vendors and consulting firms without producing comparable local capability. Domestic gains depend on whether organizations can adapt systems, develop expertise and negotiate the terms of access.

This route is especially visible in large middle-income economies with substantial consumer, platform, industrial or financial markets. Brazil, Mexico, Indonesia and South Africa illustrate different variants.

In Brazil, corporate demand has encouraged major cloud and AI investment. Microsoft announced a three-year investment of 14.7 billion reais to expand cloud and AI infrastructure in

São Paulo state and train 5 million people in AI-related skills [1]. The case shows how market demand and infrastructure expansion can reinforce one another.

Mexico's trajectory is tied more closely to manufacturing, nearshoring and integration into North American supply chains. In October 2025, Salesforce announced a five-year investment of US$1 billion to expand its operations and establish a Global Delivery Center serving clients across the Americas [2]. AI enters here as part of a broader transformation of industrial and business services.

Indonesia provides a platform-market variant. Southeast Asia's largest digital economy has extensive e-commerce, digital finance, transport, delivery and online media platforms that generate demand for recommendation systems, fraud detection, customer service and logistics optimization [3, 4]. This market-led adoption has developed alongside the National AI Strategy 2020–2045, work on an updated roadmap and the expansion of cloud infrastructure, including the AWS Jakarta Region [5–7]. Consumer platforms and business processes therefore made AI visible at scale before compute, research capability and state coordination formed a connected national system.

In South Africa, Huawei Cloud reported serving more than 1,000 customers across government, finance, education, telecommunications, media and e-commerce after five years of local operation [8]. The breadth of this customer base indicates that cloud-based AI demand already spans several institutional fields, even without a coordinated national capacity strategy.

Market entry exposes a central tension: use can expand faster than bargaining power. Adoption does not reveal who controls models, platforms, data pipelines, contracts or technical standards. A market becomes a foundation for capacity only when demand supports local adaptation, expertise, industrial upgrading and public capability.

1.2. Entry through infrastructures: AI as cloud, data centres and compute

Infrastructure provides a second route. AI becomes consequential through the material systems that support computation: cloud regions, data centres, high-performance computing, storage and connectivity. These investments draw AI into energy planning, urban development, land use, environmental governance and regional strategy.

Compute-intensive AI has sharpened the importance of this route. For countries outside the main centres of model development, the decisive issue is not the territorial presence of data centres alone. Ownership, workloads, access conditions and the ability of universities, firms and public agencies to use compute determine whether infrastructure strengthens national or regional capacity.

South Africa offers one of the clearest African examples. AWS opened its Africa (Cape Town) Region in April 2020 [9], and Google Cloud made its Johannesburg region available in January 2024 [10]. Huawei Cloud had established a local hyperscale data centre in 2019 and later linked rising AI adoption to increased corporate cloud spending [8]. Together, these investments position South Africa as an infrastructure location serving a wider African market.

Mexico represents a different geography of infrastructure entry. In September 2025, CloudHQ announced a US$4.8 billion project to build six data centres in Querétaro, linked explicitly to cloud and AI workloads and designed to use waterless cooling [11]. The project connects AI infrastructure to nearshoring, proximity to the United States and the availability of land, grid capacity and network connectivity.

Indonesia's infrastructure trajectory combines data centres, subsea connectivity and sovereign investment. In September 2025, the Indonesia Investment Authority identified data centres and digital infrastructure as priorities, supported the DayOne project in Batam and entered a partnership with Granite Asia to invest more than US$1.2 billion in the country's technology and AI ecosystem [12]. In an archipelagic state, AI infrastructure is closely tied to connectivity, data control and investment strategy.

The Gulf states illustrate infrastructure entry through sovereign capital and geopolitical positioning. Microsoft's US$1.5 billion investment in UAE-based G42 formed part of a partnership to expand AI and digital infrastructure across the Middle East, Central Asia and Africa [13]. Here infrastructure investment serves not only domestic provision but also an attempt to secure a strategic position in the wider AI economy.

Infrastructure can therefore become consequential before a country develops major models or applications. Territory, electricity, cooling, connectivity, ownership and cloud contracts shape the terms of participation. Yet local facilities do not guarantee local control: a review of 775 data-centre projects outside the United States found that foreign operation can limit the sovereignty gains associated with domestic hosting [14]. The relevant test is the access, control and bargaining power that infrastructure creates.

1.3. Entry through state systems: AI as the next layer of digital government

State systems provide a third route. Governments and public agencies add AI to existing digital infrastructures, including identity, payment, registry, document and service-delivery systems. This makes AI a question of public capacity: whether institutions can define needs, govern data, procure systems and retain control over implementation.

India is the clearest example. AI is developing on top of an established digital public infrastructure that includes Aadhaar, UPI, DigiLocker and other public platforms. Bhashini adds machine translation and speech technologies intended to widen access to digital services across more than 22 Indian languages [15]. The IndiaAI Mission adds compute, datasets, innovation support and indigenous capability; its official framework emphasizes wider access to computing, improved data quality and socially relevant applications [16].

Brazil combines state-system entry with industrial policy and public-service modernization. Its 2024-2028 AI plan, valued at 23 billion reais, allocates funding to infrastructure and training as well as applications in public health, agriculture, education, business and environmental policy [17].

Rwanda and the UAE offer further state-led variants. Rwanda's national AI policy links AI to economic growth, public-service delivery and inclusive development [18]. The UAE has embedded AI in smart-government programmes, public-sector transformation and sovereign technology companies. In both cases, state institutions define AI as an instrument of administrative and developmental strategy.

State-system entry shifts governance toward procurement, public data, institutional mandates and administrative expertise. Vendor selection, contract design, access to documentation and responsibility for maintenance can shape public capacity long after a system is introduced.

1.4. Entry through resource and industrial regimes: AI as optimization, extraction and productivity layer

Resource and industrial regimes provide a fourth route. AI becomes economically significant where firms and public utilities already possess capital-intensive assets, operational data and strong incentives to improve performance. It enters as a tool for prediction, maintenance, automation, safety and resource management rather than as a stand-alone technology sector.

This route directs attention to the sectors in which many Global South economies already create value. Mining, agribusiness, manufacturing, logistics, energy and maritime systems can become early sites of AI use because they combine complex operations with large volumes of domain-specific data.

The configuration differs by country. South Africa connects AI to mining, energy and logistics as well as finance and cloud services. Brazil's trajectory includes agribusiness and industrial systems, while Mexico's manufacturing base creates demand for automation and supply-chain management. Indonesia's commodities and maritime geography give logistics and resource operations particular importance.

The central issue is how AI changes the distribution of value and control inside these sectors. Productivity gains may leave ownership of data, software and technical expertise unchanged, reinforcing established patterns of concentration. Capacity formation requires stronger local engineering capability, institutional oversight and a wider distribution of benefits.

This route also links industrial use to the resource demands of AI itself. Data centres draw on electricity, water, land and grid capacity, while industrial applications require sensors, connectivity and sector-specific expertise. AI can therefore optimize resource production while simultaneously placing new demands on the systems that provide it.

1.5. Entry through language ecologies: AI as linguistic infrastructure

Language provides a fifth route. In multilingual societies, participation depends on whether AI systems can process the languages through which people access services, express knowledge and interact with institutions. Corpora, speech datasets, translation tools and local models form part of the infrastructure of access.

Global AI systems remain uneven across languages. The AI4D African Language Program notes that advanced speech and language technologies are concentrated in high-resource languages, while many African languages lack foundational datasets and baseline models. The programme reports producing more than nine open-source datasets and related baseline models [19].

India's linguistic scale makes this route especially important. Bhashini aims to reduce language barriers in digital services [15], while AI4Bharat develops open-source datasets, tools and models for Indian languages, including speech recognition and machine translation [20].

African countries face related but distinct challenges. Nigeria, Kenya, South Africa, Ghana and Ethiopia combine large multilingual populations with uneven digital coverage. South Africa now recognizes twelve official languages, making multilingual AI a direct public-service issue [21]. Ghana has also become an important research node: Google opened its first African AI research centre in Accra in 2018 [22], and the centre later contributed to the Open Buildings

Dataset through machine-learning analysis of satellite imagery [23]. These examples show how language capacity can intersect with public data, geography and applied research.

Language entry broadens the meaning of AI capacity. It concerns the ability to make local languages, administrative vocabularies and everyday forms of communication technically usable, while retaining influence over the datasets and models through which that visibility is produced.

1.6. Entry through development agendas: AI as problem-solving ecosystem

Development agendas provide a sixth route. Universities, donors, international organizations, civic groups and local laboratories introduce AI through work on health, agriculture, education, climate adaptation and public services. This route frames AI around problem-solving and social value.

The route is especially visible in Africa. AI4D Africa was established in 2020 with a CAD 20 million investment from IDRC and Sida to support locally led research, policy hubs and multidisciplinary laboratories [24]. The wider programme reports £60 million in commitments, activity in more than 36 countries and 13 multidisciplinary labs [25].

Kenya, Ghana, Rwanda, Nigeria and South Africa illustrate different development-oriented ecosystems. Their trajectories combine local research, public-service experimentation, mobile and platform economies, responsible-AI networks and sector-specific applications. Rwanda's policy explicitly connects this activity to public services and inclusive development [18].

Development-oriented entry can connect AI to local needs at an early stage, but it also creates a risk of projectization. Pilots and research networks may remain dependent on short funding cycles and external agendas unless public institutions acquire budgets, procurement routes and responsibility for long-term operation.

1.7. A typology of primary AI entry points

The six entry points are summarized below.

Table 1.7. A typology of primary AI entry points

Primary entry point	What AI builds on first	Primary examples	First actors pulled into the AI field	Early capacity possibility	Early dependency risk
Markets	Enterprise software, finance, platforms, consumer services, corporate demand	Brazil, Mexico, Indonesia, South Africa	Cloud vendors, banks, telecoms, retailers, consultants, large firms	Local adaptation, enterprise capability, sectoral productivity	Imported platforms, API dependence, vendor lock-in
Infrastructures	Cloud regions, data centres, connectivity, GPUs, compute contracts	South Africa, Mexico, Indonesia, UAE, Saudi Arabia, Chile	Hyperscalers, data-centre firms, energy utilities, investors, municipalities	Regional compute access, infrastructure bargaining power	Foreign-operated infrastructure, limited local access, resource pressure

State systems	Digital identity, payments, registries, public platforms, procurement	India, Brazil, Rwanda, UAE, South Africa	Digital government agencies, ministries, public procurement bodies, regulators	Public AI capacity, service delivery, data coordination	Procurement dependency, weak accountability, fragmented mandates
Resource and industrial regimes	Mining, agriculture, logistics, energy, manufacturing, ports, utilities	South Africa, Brazil, Mexico, Indonesia, Saudi Arabia, Chile	Industrial firms, resource companies, utilities, engineering providers	Industrial upgrading, operational capacity, sectoral innovation	Extraction, surveillance, productivity gains without local capability
Language ecologies	Multilingual societies, low-resource languages, speech and text datasets	India, Nigeria, Kenya, South Africa, Ghana, Ethiopia	Universities, language communities, open-source groups, public service agencies	Local language models, inclusion, accessible public services	Data extraction, uneven language coverage, dependence on external models
Development agendas	Health, agriculture, education, climate, inclusion, responsible AI networks	Kenya, Ghana, Rwanda, Nigeria, South Africa, India	Donors, universities, NGOs, civic tech, local labs, policy hubs	Locally relevant AI, research networks, social problem-solving	Projectization, external agenda-setting, weak institutionalization

The typology identifies primary routes, not fixed country categories. Trajectories become layered as AI spreads into adjacent domains. India begins from digital public infrastructure and adds compute, language technologies and innovation support. South Africa combines corporate demand with cloud and data-centre investment, while Brazil links markets, industry and public investment. Mexico connects industrial integration with data-centre geography; Kenya and Ghana show how research and development networks can precede large-scale compute; and the UAE and Saudi Arabia use infrastructure finance and sovereign capital to pursue regional position.

AI diffusion therefore follows the institutional and economic conditions already present in a country. It develops through existing systems of production, governance and social organization rather than creating a trajectory from scratch.

The primary entry point shapes the initial political economy of AI: who captures value, who sets technical and contractual terms, where governance responsibility falls and which dependencies become difficult to reverse.

The analysis must therefore move from entry to capacity formation. The spread of AI matters insofar as it strengthens the ability to access, build, adapt, govern and sustain the infrastructures, resources, data, skills and institutions required for meaningful participation in AI development and use.

2. Why entry points matter

This paper uses entry points to move from the visible spread of AI to the terms on which countries enter the AI economy. Adoption indicators record use, but they do not establish who controls infrastructure, data, models or contractual conditions [26, 30, 34].

This distinction is particularly important in the Global South, where access to the assets required for AI remains uneven. UNCTAD identifies infrastructure, data and skills as the three principal leverage points for inclusive AI development [26]. The market for AI compute is concentrated among a small number of hardware, software and cloud providers [34], while the 2026 AI Index reports private AI investment of USD 285.9 billion in the United States in 2025, compared with USD 12.4 billion in China [27]. The framework developed here treats the route of entry as one factor shaping participation in this unequal economy.

2.1. Entry points show where AI becomes consequential first

AI does not become consequential only when a government adopts a national strategy or creates a dedicated regulatory body. Its effects can emerge earlier, in the first domain where AI-related decisions begin to alter investment priorities, institutional responsibilities or access to services.

The location of this initial change varies. Corporate and platform-led adoption can strengthen the influence of firms that already control digital infrastructures and occupy privileged positions in policy processes [30, 33]. The expansion of data centres draws electricity, water, land and compute allocation into decisions about AI development [28, 34]. In the public sector, the introduction of AI systems gives procurement authorities and user agencies a direct role in determining contractual control, administrative capability and accountability [31, 32].

These decisions can shape the conditions of AI development before the state formally recognizes AI as a distinct policy field. Procurement and data-licensing agreements may establish long-term dependence on particular vendors or proprietary systems [31, 32]. Infrastructure investment can commit land, energy and connectivity to specific projects and operators well before a national AI strategy defines broader priorities [28, 34]. By the time formal coordination begins, some of the most consequential choices may therefore already have been made.

This paper uses the term entry point to identify the domain in which this process becomes visible first. The aim is to trace where AI begins to redistribute resources, expertise and decision-making authority, and which actors acquire influence at that stage. This does not imply that national strategies are unimportant. It means that they may enter a field whose institutional and material conditions have already begun to take shape.

2.2. Entry points show which actors shape the agenda

Different routes give early influence to different coalitions. Research on Big Tech shows how control over platforms, infrastructure and technical expertise can translate into agenda-setting influence [33]. Studies of public AI procurement likewise show that vendors, purchasing bodies and user agencies shape regulatory outcomes through contracts and standards [31, 32]. The actor categories used here synthesize these findings with the country evidence presented in Section 1.

Table 2.2. Entry points show which actors shape the agenda

Entry point	Actors with early influence	Dominant framing	Initial governance problem

Markets	Cloud providers, software vendors, banks, telecoms and large firms	Efficiency, automation and competitiveness	Vendor dependence, data protection, competition and labour effects
Infrastructures	Hyperscalers, data-centre developers, utilities, investors and municipalities	Compute, connectivity and infrastructure investment	Energy, water, land, ownership and local access
State systems	Ministries, digital agencies, procurement bodies and regulators	Service delivery and administrative modernization	Procurement, accountability, data governance and expertise
Resource and industrial regimes	Industrial firms, utilities, logistics companies and engineering providers	Optimization, safety and productivity	Data control, labour, environmental effects and value distribution
Language ecologies	Universities, language communities and open-source initiatives	Representation, inclusion and access	Control of language resources and uneven coverage
Development agendas	Donors, universities, NGOs, research labs and policy networks	Public value and social problem-solving	Project dependence and weak institutionalization

Source: author's synthesis based on the country evidence in Section 1 and [31-33].

AI governance may therefore be distributed across institutions that control procurement, infrastructure, data access and research funding even when another body leads the national strategy [31-33].

2.3. Entry points show which institutions are pulled into governance

AI can turn institutions without a formal AI mandate into de facto governance actors. Public procurement research shows that purchasing bodies and user agencies shape accountability, contractual control and dependence on technology vendors [31, 32]. Research on platform power likewise demonstrates that competition and regulatory authorities increasingly deal with firms whose influence cuts across established policy domains [30, 33].

An infrastructure entry point broadens this field further. Data centres depend on grid connections, land allocation, cooling systems and environmental permits, which brings energy regulators, environmental agencies and municipalities directly into the governance of AI infrastructure [28, 34]. The IEA estimates that data-centre electricity consumption reached about 485 TWh in 2025 and could approach 950 TWh by 2030 [28].

These arrangements create a recurrent institutional problem. The body formally assigned responsibility for AI policy may set general principles or coordinate a national strategy, while decisions that determine access, dependence and resource use are taken elsewhere: by procurement authorities, sectoral regulators, infrastructure agencies or local governments. This paper refers to that gap as governance misalignment. It arises when responsibility for AI is formally concentrated in one part of the state, but the authority, budgets and technical expertise needed to shape its development are distributed across institutions that may not share the same objectives or accountability mechanisms. Evidence from public-sector AI procurement illustrates how fragmented oversight and uneven institutional capability can produce such a gap [31, 32].

2.4. Entry points show which dependencies form early

The domain through which AI enters can shape the dependencies that emerge at an early stage. Technical and contractual choices matter because they determine which infrastructures, suppliers and standards institutions must rely on, and how easily these arrangements can later be changed.

Cloud services provide one example. OECD research identifies high market concentration, limited interoperability and substantial switching costs in cloud markets [30]. Similar constraints arise in public-sector AI procurement when proprietary technologies, restrictive licences or weak portability provisions tie agencies to particular vendors and make data or systems difficult to transfer [31, 32]. Concentration in compute markets narrows the range of alternatives further, especially for institutions that cannot finance or operate their own infrastructure [34].

The form of dependence also varies with the entry point. Platform-led adoption can place data, technical expertise and access to policymakers in the hands of large firms [33]. Language initiatives may rely on datasets, models and computational resources that remain outside the control of the institutions expected to use or maintain them [19, 20]. Development programmes face a different problem: externally funded pilots may create temporary access to expertise and infrastructure without establishing the budgets, organizational routines or institutional ownership required for continuation. Research on public-sector innovation shows that scaling frequently encounters structural, organizational and stakeholder barriers [35].

This paper uses the term “initial architecture of dependence” to describe the way these early choices accumulate. A procurement contract may create reliance on one supplier; the selected system may require a particular cloud environment; limited local compute may then make switching even harder. Dependence therefore does not arise from a single decision. It develops through the interaction of contractual, infrastructural and institutional constraints. The concept directs attention to what remains outside local control, which alternatives disappear as a result, and which decisions become increasingly costly to reverse [30–35].

2.5. Entry points show what kind of capacity may or may not emerge

Entry points matter not only because they create early dependencies, but also because they channel investment, expertise and institutional attention towards particular capabilities. An infrastructure-led trajectory initially privileges compute, connectivity and resource coordination. Public-sector adoption places greater weight on procurement, oversight and administrative capability. Language initiatives foreground data stewardship, model development and the institutional maintenance of linguistic resources. These categories synthesize the evidence reviewed in Section 1 with broader work on AI infrastructure, compute capacity and public-sector innovation [26, 34, 35].

The resulting capacity is rarely complete. A country may strengthen one component while leaving the connections to other components weak. Compute infrastructure may expand without widening effective access; technical expertise may grow without stable institutional demand; a successful pilot may remain detached from routine budgets and administrative processes [30–35]. The entry point therefore helps explain which capabilities appear first, but not whether they will combine into a durable system.

In this paper, entry points are treated as formative conditions. They influence the sequence of capacity formation by directing resources and authority towards some functions earlier than others. The later outcome depends on whether institutions can connect these initially

uneven capabilities across infrastructure, data, expertise, governance and developmental priorities.

2.6. Entry points and the limits of readiness thinking

The entry-point approach also helps move beyond readiness rankings. Such indices can be useful for broad comparison and for identifying gaps in infrastructure, data, governance, skills and innovation systems. The 2025 Government AI Readiness Index, for example, assesses 195 governments according to their capacity to harness AI for public benefit [29]. Its comparative logic is valuable, but it differs from the question posed here. Entry-point analysis asks how a country is being incorporated into the AI economy, which actors and institutions shape that process, and what the route of incorporation makes possible or difficult.

Countries with comparable readiness scores may therefore have arrived there through very different configurations of infrastructure, ownership, public authority and access. This is an inference generated by the framework proposed here, not a conclusion drawn from the index itself.

The two approaches are therefore complementary. Readiness indices identify the conditions present at a given moment; entry-point analysis reconstructs the trajectory through which those conditions were assembled and the constraints embedded in that process.

2.7. From entry point to trajectory

The main value of the entry-point concept is that it turns AI diffusion into a trajectory. AI enters through one route, then begins to reorganize other domains.

The route of entry shapes what follows. Market concentration and platform control can stimulate demand while preserving external dependence [30, 33]. Infrastructure investment can expand compute supply but leave questions of access and allocation unresolved [34]. Public-sector deployment can expose weaknesses in procurement and accountability [31, 32], while pilots require organizational strategies to move into routine systems [35].

The trajectory is therefore not determined by the entry point, but the entry point sets the initial direction. It shapes the early institutional grammar of AI: what counts as a problem, who is considered an expert, which resources are mobilized, which risks are prioritized, and what kind of capacity is imagined.

This is why identifying entry points matters for the broader argument of this paper. If AI begins differently across countries, then AI governance in the Global South cannot be built only around common regulatory templates, abstract principles or readiness scores. It needs an account of capacity formation: how AI becomes embedded in existing markets, infrastructures, state systems, resource regimes, language ecologies and development agendas, and whether this embedding creates durable capacity or new dependency.

2.8. Diagnostic summary

The diagnostic contribution of entry points can be summarized through five questions. The table is an authorial synthesis based on the evidence and literature cited in Sections 1 and 2.

Table 2.8 Diagnostic contribution of entry points

Diagnostic question	What it reveals	Why it matters
Where does AI become consequential first?	The first domain in which AI changes priorities, investment or institutional authority	Identifies the initial policy problem and site of governance
Who shapes the agenda?	The actors that define relevant applications, risks and strategic assets	Locates early agenda-setting power
Which institutions enter governance?	The organizations that manage AI-related consequences, often without a formal AI mandate	Shows where coordination and authority are fragmented
Which dependencies form early?	Contractual, infrastructural, technical, data, language, vendor or donor dependence	Identifies choices that may become costly to reverse
What capacity can emerge?	The first organizational, infrastructural, public, sectoral, linguistic or developmental capabilities	Shows whether diffusion can produce durable capacity

The term "AI adoption" can refer to materially different processes, including the purchase of a cloud service, approval of a data centre, deployment of a public-sector system, creation of a language dataset or launch of a funded pilot [26, 28, 31, 32, 35]. Identifying the entry point makes these differences explicit and provides a basis for tracing changes in authority, resources and institutional possibilities.

3. From AI diffusion to AI capacity

The entry-point typology explains how AI first becomes consequential. It does not establish whether diffusion leaves countries with greater control over the systems they use. Research on organizational AI capability and public administration distinguishes adoption from the complementary resources and routines needed to develop, evaluate and sustain AI [36, 41]. Studies of the compute divide likewise show that widespread use can coexist with concentrated control over the computational resources required for advanced research and scrutiny [38, 39].

This paper uses AI capacity to examine that difference. The concept shifts attention from the presence of applications, strategies and infrastructure to the ability of institutions within a country or region to influence how AI is accessed, adapted, governed and maintained. The relevant outcomes include practical access to compute, usable data and language resources, institutional competence, resource governance and alignment with locally defined development priorities. These dimensions draw on existing work on organizational capability, compute access, policy capacity and inclusive AI development [26, 34, 36–41, 47].

3.1. Defining AI capacity

Organizational studies define AI capability as the capacity to mobilize complementary technological, data, human and organizational resources rather than merely possess a technology [36]. Research on big-data analytics reaches a similar conclusion: infrastructure and data create value when organizations connect them to managerial expertise and established routines [37]. At a wider scale, the compute-divide literature demonstrates how unequal access to computational resources concentrates research and model scrutiny in large firms and a small group of well-

resourced institutions [38, 39]. Compute-governance research further shows that ownership and allocation affect access and control, turning computational infrastructure into a political as well as technical resource [40].

These literatures provide the foundations for a broader political-economic concept. This paper defines AI capacity as the ability of a country or region to access, build, adapt, govern and sustain the infrastructures, resources, data, skills and institutions required for meaningful participation in AI development and use. In condensed form:

AI capacity is the ability to shape the material, institutional and knowledge conditions under which AI is developed, accessed and used.

The definition treats meaningful participation as more than consumption. It requires enough capability to adapt systems to local contexts, evaluate their effects, influence procurement and access conditions, and direct their use toward locally defined priorities. Organizational capability research, public-administration studies and the compute-divide literature all support the underlying distinction between use and effective control [36, 38, 39, 41]. The definition itself, however, is the analytical contribution proposed in this paper.

Bargaining power forms part of this capacity. Governments and public institutions can exercise it through procurement terms, compute-access arrangements, data rules, interoperability standards and conditions attached to public finance or infrastructure approvals. Existing research documents vendor and data lock-in in public procurement, concentration in cloud and compute markets, and the role of standards and portability in limiting dependence [30-34]. This paper therefore treats bargaining power as the practical ability to influence the terms on which AI systems and infrastructures enter national and regional development.

AI capacity does not imply technological self-sufficiency. Advanced chips, cloud infrastructure and frontier-model development remain highly concentrated, making full autonomy unrealistic for most countries [27, 30, 34]. Capacity instead concerns adaptive capability, institutional competence and negotiating leverage within interdependent technological systems.

The concept also differs from adjacent frameworks. Adoption records the introduction or use of AI; it does not demonstrate the organizational resources required to sustain that use [36, 41]. Readiness indices compare enabling conditions across countries, but their aggregate scores can obscure differences between technological preparedness, responsible governance and institutional context [42, 43]. AI governance concerns the rules, institutions and relationships through which AI is directed and held accountable [44, 45]. Digital sovereignty centres on authority and legitimate control over digital technologies [46]. AI capacity draws on all four debates while asking a narrower operational question: what can institutions in a country or region access, coordinate, adapt, govern and sustain in practice?

3.2. Five components of AI capacity

The framework distinguishes five connected components. They are analytical categories developed in this paper, grounded in research on compute, data, organizational and policy capacity, resource requirements and inclusive development [26, 28, 34, 36-41, 47].

3.2.1. Computational capacity

Computational capacity concerns effective access to the resources needed to develop, adapt, deploy and evaluate AI. Relevant resources include accelerators, cloud services, high-performance computing, storage, networking and the expertise needed to use them. The OECD notes that many national AI strategies do not adequately address domestic compute requirements, while research on the compute divide shows that unequal access constrains academic participation and independent scrutiny [34, 38, 39].

Physical location alone does not establish capacity. A country may host data centres while universities, start-ups and public agencies remain unable to obtain affordable or appropriately governed compute. OECD work on national compute capacity therefore emphasizes allocation, access, resilience and effective use alongside infrastructure provision [34]. The diagnostic question is: who can access compute, on what terms and for which purposes?

3.2.2. Data and language capacity

Data and language capacity concerns the ability to produce, document, govern, share and use data in ways that support AI development and public value. UNCTAD identifies data, infrastructure and skills as central leverage points for inclusive AI development [26]. In multilingual settings, language resources form part of this foundation because low-resource languages often lack the datasets and baseline models required for speech and language technologies [19]. India's Bhashini programme similarly treats translation and speech tools as infrastructure for access to digital services [15].

Availability is therefore insufficient. Data need documentation, interoperability, legal clarity and institutional stewardship; language resources require governance arrangements that support local research, public access and community benefit. These criteria synthesize the problems identified in the data-capacity and language-resource literature [19, 26]. The diagnostic question is: whose realities become machine-readable and usable, and under whose authority?

3.2.3. Resource and infrastructural capacity

AI systems depend on electricity, water, land, cooling, connectivity and municipal approvals. The IEA estimates that data-centre electricity demand could rise sharply by 2030, making power systems and infrastructure planning increasingly consequential for AI expansion [28]. OECD work on compute capacity also treats energy supply, connectivity and resilience as core planning requirements [34].

Resource and infrastructural capacity therefore refers to the ability to integrate AI infrastructure with energy planning, water governance, environmental regulation and spatial development. This is an analytical extension of the documented material requirements of compute infrastructure [28, 34]. The diagnostic question is: which resource systems support AI, who controls their allocation and who bears the associated costs?

3.2.4. Institutional capacity

Policy-capacity research identifies analytical, operational and political competences as necessary for effective public action [47]. Studies of AI in public administration add that the ability to develop or procure a system differs from the ability to implement, manage and sustain

it [41]. AI governance research also shows that responsibility is distributed across organizations and policy domains rather than contained within a single regulator [44, 45].

Institutional AI capacity therefore includes problem definition, procurement, vendor assessment, data governance, technical expertise, evaluation, accountability and coordination across agencies. These functions derive from the cited policy-capacity and public-governance literature [41, 44, 45, 47]. The diagnostic question is: which institutions possess the mandate, competence and resources to coordinate AI across sectors?

3.2.5. Developmental capacity

Developmental capacity concerns the ability to connect AI with locally defined economic and social priorities. UNCTAD frames inclusive AI development around infrastructure, data, skills, innovation systems and strategic positioning [26]. This paper extends that approach by asking whether AI strengthens public services, local research, productive capabilities and institutional learning, rather than merely expanding markets for imported systems.

The resulting assessment cannot rely on adoption rates alone. It must examine who gains capability, which sectors and regions benefit, and whether institutions can direct AI toward public purposes. These are analytical criteria proposed here on the basis of the inclusive-development framework [26]. The diagnostic question is: can AI activity contribute to development priorities on terms that local institutions can shape?

3.2.6. Accountability as a condition of developmental capacity

State-led capacity building can improve public administration while also increasing the state's ability to classify, predict and monitor populations. Research on public-sector AI documents risks involving opacity, weak contestability, proprietary systems and institutional dependence [50]. The UN Special Rapporteur's analysis of digital welfare systems further shows how automated administration can intensify surveillance and restrict access to effective redress [51].

Developmental capacity therefore requires accountability capacity: legal limits, documentation, audit rights, appeal procedures, human oversight and public disclosure. UNESCO's Recommendation on the Ethics of Artificial Intelligence and the OECD AI Principles place human rights, transparency, fairness, accountability and human oversight at the centre of trustworthy AI governance [48, 49]. The World Bank likewise stresses risk assessment, contestability, interoperability and institutional learning in public-sector AI [50].

Developmental capacity without accountability can become administrative power without public control.

3.3. AI capacity as a capacity to connect

The five components do not operate as a checklist of assets. Research on organizational capability shows that effective capacity emerges from combinations of complementary resources and routines [36, 37]. Policy-capacity research reaches a comparable conclusion at the institutional level: competences matter through the organizational arrangements that mobilize them [47]. This paper translates that relational insight to the national and regional political economy of AI.

A data centre does not create broad computational capacity when local institutions cannot access its services. Training programmes do not create institutional capacity when organizations cannot retain and deploy expertise. Data remain unusable without documentation and governance, while pilots fail to become systems without budgets, procurement routes and institutional ownership. Each example reflects problems documented in the relevant compute, organizational, data and public-sector literature [19, 26, 32, 34, 35, 38, 39, 41].

AI capacity is a capacity to connect.

The concept refers to the ability to connect compute with public purpose, data with usability and rights, infrastructure with resource governance, skills with institutions, and adoption with development strategy. This formulation is the paper's central analytical proposition. It draws on the relational understanding of capability in organizational and policy-capacity research but applies it to the institutional and material conditions of national and regional AI development [36, 37, 47].

Entry points shape which connections become possible first. Market-led diffusion initially strengthens organizational adoption; infrastructure-led trajectories foreground compute access and resource planning; state-system entry increases the importance of procurement and accountability; language initiatives depend on routes into public services and research; development projects require institutional ownership and durable finance. Section 1 provides the empirical basis for these distinctions [1–25]. The proposition that they form different capacity trajectories is the comparative inference developed in this paper.

3.4. A diagnostic framework for AI capacity

Table 3.4 summarizes the five-component framework. The categories and diagnostic questions are the author's synthesis; the evidence base comes from the literatures cited throughout this section [19, 26, 28, 34, 36–51].

Table 3.4 Five-component framework

Component of AI capacity	What it includes	Main diagnostic question	Typical risk if disconnected
Computational capacity	Compute, cloud, storage, networking and effective technical access	Who can access compute, on what terms and for which purposes?	Infrastructure without affordable or governable access
Data and language capacity	Governed datasets, interoperability, language resources and stewardship	Whose realities become machine-readable and under whose authority?	Unusable data, language exclusion and extraction
Resource and infrastructural capacity	Electricity, water, land, cooling, connectivity and planning	Which resource systems support AI, and who bears the costs?	Infrastructure expansion without resource accountability
Institutional capacity	Coordination, procurement, expertise, evaluation and accountability	Which institutions can coordinate and govern AI across sectors?	Strategies without implementation and vendor dependence
Developmental capacity	Public value, productive capability, inclusion and institutional learning	Can local institutions direct AI toward development priorities?	Adoption without structural or public capability

3.5. From capacity components to capacity formation

The components rarely develop at the same pace. Research on compute access, public-sector capability and innovation scaling documents recurrent gaps between infrastructure and access, technical experimentation and institutional implementation, and innovation projects and durable systems [34, 35, 38, 39, 41]. The framework developed here uses those gaps to examine how AI activity becomes (or fails to become) capacity.

The next section specifies five points of failure: adoption without control, infrastructure without access, data without usability, skills without institutions, and pilots without systems. These categories are the paper's diagnostic synthesis of the evidence reviewed in Sections 1-3.

4. Where capacity breaks: Five diagnostic gaps

The framework developed in this paper treats AI capacity as the ability to connect resources, institutions and public purposes. Existing research shows that AI adoption can advance while public organizations remain constrained by vendor dependence, limited compute access, weak data governance, fragmented expertise and difficulties in moving projects beyond the pilot stage [26, 30–35, 41]. The five gaps below organize these recurring problems into a diagnostic framework. They are the paper's analytical synthesis rather than a universal sequence through which every country must pass.

The framework distinguishes adoption without control, infrastructure without access, data without usability, skills without institutions, and pilots without systems. Each gap identifies a failed conversion: use does not become influence, territorial infrastructure does not become accessible compute, records do not become governed data, individual expertise does not become organizational capability, and experimentation does not become routine institutional practice.

4.1. Adoption without control

Adoption without control describes situations in which organizations use AI systems but cannot independently evaluate, modify, audit or replace them. Public-sector procurement is especially important because contractual terms determine access to documentation, data, model updates, audit rights, portability and exit options. Research on AI procurement shows that proprietary technologies and restrictive data-licensing arrangements can create vendor and data lock-in, while weak public-sector expertise limits the ability of contracting authorities to assess suppliers and maintain effective oversight [31, 32]. Studies of public administration also distinguish the ability to acquire or prototype an AI system from the organizational capability required to implement and govern it over time [41].

Control is therefore relational rather than absolute. An agency may understand a system technically but remain unable to audit it under the contract; another may possess formal audit rights but lack staff able to use them. The diagnostic test proposed here is whether the adopting organization retains practical influence over evaluation, adaptation, data use and continued operation after deployment.

Diagnostic question: *Can the organization evaluate, negotiate, adapt and govern the systems it adopts?*

4.2. Infrastructure without access

Infrastructure without access arises when cloud regions, data centres or computing facilities are located within a country but remain inaccessible to many domestic research institutions, public agencies and smaller firms. The OECD documents high concentration, substantial capital requirements and switching barriers in cloud and AI-compute markets [30, 34]. These conditions mean that territorial hosting alone does not establish who receives compute, at what price, under which contractual terms or for which purposes.

The distinction also has a material dimension. The IEA estimates that data-centre electricity consumption reached about 485 TWh in 2025 and could approach 950 TWh by 2030 [28]. Host territories therefore provide energy, land, connectivity and, depending on cooling design, water and related infrastructure. The analytical issue is whether public authorities connect these local resource commitments to affordable research access, public-sector use, skills formation or wider industrial benefits. Such connections must be demonstrated empirically; they cannot be inferred from the presence of a data centre.

Diagnostic question: *Who can use locally hosted compute, on what terms, and what public or developmental value does that access create?*

4.3. Data without usability

Data without usability describes the gap between the existence of digital records and the ability to use them responsibly for AI development or public decision-making. The World Bank's 2025 framework treats contextual data and applications as one of the foundations of AI readiness in low- and middle-income countries, alongside connectivity, compute and skills [52]. UNCTAD also identifies data as a central leverage point for inclusive AI development [26]. Neither framework equates data volume with usable capacity.

Usability depends on technical, legal and organizational conditions. Public-sector interoperability requires semantic alignment, legal authority, institutional coordination and shared standards, not only compatible software [53]. Language coverage adds another constraint: the AI4D African Language Program documents the scarcity of foundational datasets and baseline models for many African languages, while Bhashini addresses access to Indian digital services through multilingual language technologies [19, 15]. These sources support a narrower diagnostic claim: records and language resources become capacity only when institutions can document, govern, interpret and maintain them for legitimate purposes.

Diagnostic question: *Can institutions turn available records and language resources into governed, interoperable and contextually usable data?*

4.4. Skills without institutions

Skills without institutions occurs when expertise remains concentrated in individuals or temporary teams and does not become reproducible organizational capability. Research on AI and analytics capability shows that technical knowledge produces sustained value only when organizations combine it with complementary data, managerial and operational resources [36, 37]. Public-sector studies reach a similar conclusion: implementation depends on organizational arrangements, mandates and routines that connect technical expertise to the work of public

institutions [41]. The policy-capacity literature also locates effective capability across individual, organizational and system levels rather than in training alone [47].

The framework therefore does not treat graduate numbers or short courses as sufficient evidence of capacity. The relevant evidence concerns retention, career structures, access to compute and data, coordination with domain specialists, stable programme ownership and the repeated use of expertise across projects. These indicators allow researchers to test whether training investments are becoming institutional capability rather than assume that they do.

Diagnostic question: *Can organizations retain, coordinate and repeatedly deploy technical and domain expertise?*

4.5. Pilots without systems

Pilots without systems describes projects that demonstrate technical feasibility but do not become routine services or durable organizational capabilities. OECD evidence shows that many government AI initiatives remain in the pilot phase because agencies lack impact-assessment frameworks, implementation guidance, skills and access to quality data [32]. Research on scaling public-sector innovation similarly identifies governance, funding, evaluation and organizational integration as conditions for moving beyond isolated experiments [35].

A successful demonstration does not establish that an institution can finance, procure, maintain, monitor and contest the system under ordinary conditions. The diagnostic framework therefore looks for a responsible owner, a continuing budget, integration into workflows, data-maintenance arrangements, user support, evaluation procedures and a route for accountability. These are observable conditions; the label "successful pilot" should not substitute for evidence that they exist.

Diagnostic question: *Can the experiment be financed, governed, maintained and incorporated into ordinary practice?*

4.6. The five gaps as a diagnostic framework

The table 4.6 converts the five gaps into observable research questions. It does not assign a score or assume that all gaps carry equal weight. Researchers can use it to identify where a specific AI initiative or national trajectory fails to convert activity into durable capacity.

Table 4.6 The five gaps as a diagnostic framework

Diagnostic gap	Failed conversion	Evidence to examine	Core question	Typical dependency risk
Adoption without control	Use → practical influence	Contracts, audit rights, documentation, portability, internal evaluation capability	Can adopters evaluate, adapt and govern the system?	Vendor and data lock-in
Infrastructure without access	Hosting → usable compute	Pricing, allocation, research access, public-sector use, investment conditions	Who can access compute and on what terms?	Foreign-operated infrastructure with narrow domestic access

Data without usability	Records → governed data	Metadata, standards, legal basis, interoperability, stewardship, language coverage	Can institutions use data legitimately and reliably?	Fragmented records, extraction and exclusion
Skills without institutions	Expertise → organizational capability	Retention, careers, teams, compute, coordination, programme ownership	Can organizations sustain and reproduce expertise?	Dependence on consultants or temporary teams
Pilots without systems	Experiment → routine practice	Budget, owner, procurement, maintenance, workflow integration, evaluation and redress	Can the project operate beyond exceptional pilot conditions?	Projectization and repeated experimentation

The gaps can reinforce one another, but the direction and strength of these relationships require empirical investigation. Vendor dependence may limit access to documentation; weak data governance may impede implementation; temporary staffing may leave systems without maintenance; and inaccessible compute may narrow the range of actors able to develop or evaluate applications [30–35, 41]. The framework treats these as testable connections rather than automatic outcomes.

4.7. From diagnostic gaps to empirical investigation

The diagnostic gaps are intended as questions for empirical investigation, not as categories for ranking countries or assigning fixed capacity profiles. The same apparent gap may emerge through different institutional routes, while similar entry points can produce different outcomes depending on contractual arrangements, regulatory authority, organizational capability and access to resources. A market-led trajectory may therefore direct attention to ownership and value capture; an infrastructure-led trajectory to effective access, resource commitments and local oversight; and a state-system trajectory to procurement, data governance and implementation capability. These are analytical propositions generated by the framework. Establishing whether a gap exists requires country- and sector-specific evidence.

The next section applies this logic to India and South Africa in order to show why commonly used indicators are insufficient for assessing AI capacity. The comparison therefore has a methodological purpose. It identifies the claims that available evidence can support, the points at which those claims become speculative, and the additional data needed to assess access, use, control and institutionalization. By doing so, the country applications turn the diagnostic framework into a research strategy and establish the empirical basis for the agenda developed in Section 6.

5. Applying the framework to India and South Africa: evidence and unresolved questions

5.1. India: from programme scale to questions of institutional use

India's AI trajectory develops on top of a digital public infrastructure that already connects identity verification (Aadhaar), payments (Unified Payments Interface, UPI), document exchange (DigiLocker), and access to public and commercial services. India Stack is commonly used as an umbrella term for this architecture. It brings together digital public systems and open interfaces through which government agencies, banks and private service providers can verify users, exchange authorized data and build services on shared technical rails [54].

Three components illustrate its scale and institutional reach. Aadhaar provides residents with a unique biometric identity number that can be used for authentication when accessing public benefits, banking and other services. DigiLocker allows government bodies and other authorized institutions to issue and verify digital documents [55]. UPI enables instant bank-to-bank payments through a common interoperable system rather than through a single commercial provider [57]. Together, these systems create an environment in which identity, payments and official records can be connected across a wide range of services.

Government reporting indicates the scale of this environment. By March 2026, more than 1.44 billion Aadhaar numbers had been generated, while the system processed 27.07 billion authentication transactions during 2024–25 [56]. In March 2026, UPI recorded 22.64 billion transactions worth ₹29.53 trillion across 703 participating banks [57]. These figures establish the reach and transaction volume of the digital systems on which AI-enabled services may be developed. They do not establish who controls the models, cloud services and data-processing arrangements subsequently connected to them.

The Indian case is therefore examined at two levels. The first concerns the resources that national programmes have mobilized, particularly compute and language technologies. The second concerns what the available evidence can establish about access, institutional use, maintenance and control.

5.1.1. Compute: from announced provision to demonstrated use

The IndiaAI Mission placed shared compute within a national policy programme. In March 2024, the Cabinet approved a five-year budget of ₹103.72 billion and proposed a common computing facility with at least 10,000 GPUs, alongside datasets, application development, start-up finance, indigenous models and responsible-AI measures [16]. A Press Information Bureau release published in March 2026 reported that more than 38,000 GPUs had been onboarded through the IndiaAI Compute Portal and offered to start-ups and academic institutions at subsidized rates [58].

The official figure documents resources included in the programme's provision mechanism. It is not a direct measure of allocation or use. Independent reporting in March 2026 described slow allocation, long deployment periods, financial constraints among recipient firms and underuse of some allocated GPUs [59]. The two sources therefore illuminate different stages of the programme: the government release documents formal availability, while the independent report raises questions about how quickly and effectively organizations can use the resources offered through it.

The available evidence supports a claim of substantial state mobilisation and rapid programme expansion. Assessing the programme's contribution to institutional capacity would require more systematic data on allocation times, prices actually paid, user composition,

utilization rates, technical support and the distribution of access across firms, universities, research centres and regions.

5.1.2. Language infrastructure: resources, integration and maintenance

India's linguistic diversity creates a practical challenge for digital services. Many systems operate primarily in English or in a limited number of widely used languages, while citizens may need to access welfare, health, education and administrative services in one of the country's 22 scheduled languages.

The Indian government launched BHASHINI under the National Language Translation Mission to develop a shared public platform for Indian-language technologies. It provides access to machine translation, speech recognition, speech synthesis and related language services that public agencies and other organizations can integrate into digital applications [15].

Academic initiatives such as IndicTrans2 expand this infrastructure from the technical side. Its developers assembled new parallel corpora, created benchmarks and released open models after identifying gaps in training data, India-specific evaluation and language coverage [60]. BHASHINI and IndicTrans2 therefore provide evidence of public investment, technical development and the growing availability of multilingual datasets and models.

The reviewed sources provide less information on how routinely these technologies are integrated into hospitals, schools, welfare offices or local administrations. They also offer limited evidence on performance in specialized domains, the distribution of use across languages and regions, and the institutional arrangements through which models are evaluated and maintained. Assessing language capacity would therefore require evidence of repeated use, service-level performance, organizational responsibility and long-term updating, rather than model availability alone.

5.1.3. Data-centre expansion and local resource governance

National compute programmes form only one part of India's infrastructure trajectory. The expansion of commercial data centres also depends on land, electricity, water, municipal permissions and local planning. These conditions become visible when proposed facilities encounter opposition in the places where they are to be built.

In July 2026, more than one hundred residents protested outside the site of a planned Amazon data centre in Thane. The proposed development, valued at ₹450 billion and occupying a 58-acre site, prompted claims concerning noise, environmental risks and pressure on local water and electricity supplies. Local political representatives called for project permissions to be reviewed by an independent expert committee and for a detailed public hearing. The dispute emerged while the Maharashtra government was promoting the Mumbai Metropolitan Region as a major hub for data-centre investment [61].

The source documents the protest and the claims made by residents and political representatives. It does not independently establish the facility's projected resource consumption, whether environmental rules were breached, or what effects the project would have if completed. The case therefore cannot support a conclusion about the environmental impact of the proposed data centre.

It does show that the expansion of digital infrastructure is governed through decisions made beyond national technology programmes. Municipal authorities, utilities, state agencies,

developers, political representatives and affected communities become involved through land allocation, environmental review, infrastructure provision and public consultation. The Thane dispute consequently identifies a further field for empirical analysis: whether the institutions responsible for approving data-cententre development can assess resource requirements, disclose expected impacts and reconcile national infrastructure ambitions with local planning and accountability.

5.1.4. What remains to be established

Publicly available sources document India's programme scale and formal resource provision more clearly than the institutional use and local governance of those resources. They establish the reach of digital public infrastructure, the expansion of shared compute, the creation of multilingual technologies and the emergence of contestation around data-centre development. They do not permit a systematic assessment of how these resources and associated decision-making powers are distributed across institutions, sectors and regions.

A fuller assessment would require evidence on:

- the allocation, affordability and utilization of computational resources;
- the institutional and regional distribution of access to national programmes;
- the routine integration and performance of language technologies in public services;
- the organizations responsible for maintenance, evaluation and technical support;
- procurement, licensing and cloud arrangements that determine control, portability and accountability;
- the resource assessments, approval procedures and public-consultation mechanisms applied to data-centre development.

Such evidence would allow the framework to trace where formal provision develops into sustained organizational capability and where access, expertise, institutional authority or local accountability remain limited. Until it is available, the Indian case supports conclusions about the scale and direction of mobilisation, while leaving the distribution, durability and governance of the resulting capacity open to further investigation.

5.2. South Africa: from infrastructure presence to questions of local capacity

South Africa presents a different evidentiary and institutional configuration. The country combines advanced corporate infrastructure and an established university research base with persistent economic inequality and uneven access to digital services. The OECD describes an economy constrained by low growth, very high unemployment and inequality [62], while the World Bank characterizes South Africa as a dual economy in which advanced sectors coexist with severe exclusion [63]. Bibliometric research also documents a substantial national output in artificial intelligence, concentrated in several leading universities [64].

These sources establish that South Africa possesses important technological, commercial and research assets. They provide less information on how these assets connect to public institutions, smaller firms, universities outside the leading centres and communities with limited access to reliable digital infrastructure. The case is therefore used to examine the distinction between the territorial presence of advanced infrastructure and evidence of locally usable institutional capacity.

5.2.1. Cloud infrastructure: territorial presence and local access

South Africa hosts cloud regions operated by several major global providers. Amazon Web Services opened its Africa Region in Cape Town in 2020 [9], Google Cloud opened a Johannesburg region in 2024 [10], and Huawei Cloud reported operating a local hyperscale data centre and serving more than 1,000 customers across multiple sectors [8]. These sources establish that South Africa has become a significant location for regional cloud infrastructure and commercial service provision.

Territorial presence does not by itself establish the conditions under which local institutions gain access. Corporate announcements do not provide systematic information on prices paid by universities, start-ups and public bodies, the availability of subsidized or research-oriented compute, contractual portability, audit rights or the extent of local participation in infrastructure design and operation.

A fuller assessment would therefore need to distinguish between infrastructure located in South Africa and infrastructure that expands the capabilities of South African institutions. Relevant evidence would include the terms of public cloud procurement, access programmes for researchers and smaller firms, local supplier participation, knowledge transfer, data-portability provisions and the extent to which public agencies can negotiate technical and contractual conditions.

5.2.2. Resource governance and municipal accountability

The development of cloud and data-centre infrastructure also depends on electricity, land, water, planning permissions and environmental oversight. Reuters reported that Africa Data Centres and DPA SA began constructing a 12 MW solar facility in 2024 under a twenty-year power-purchase agreement. The project linked private electricity generation to data-centre demand and concerns about grid reliability [65].

A proposed Equinix development in Cape Town generated a different type of evidence. Local actors requested information on water consumption, electricity demand, emissions, noise and backup generation. Equinix stated that it had acquired land but had not yet submitted a planning application [66]. The available reporting documents the questions raised around the project; it does not establish the facility's eventual environmental impact or whether any regulatory breach occurred.

These cases show where infrastructure governance takes place. Municipalities, utilities, provincial authorities, developers and affected communities become involved through energy provision, land-use decisions, environmental review and public participation. Assessing local capacity in this domain would require evidence on the quality of resource assessments, disclosure requirements, coordination among authorities and the ability of local institutions to connect private investment to wider infrastructure planning.

5.2.3. Connectivity and the conditions of participation

The presence of major cloud regions coexists with unequal household connectivity. Statistics South Africa reported that 75.6 per cent of households accessed the internet through mobile devices in 2024, while 17.4 per cent had fixed internet access at home. Fixed home

access reached 27.2 per cent in metropolitan areas, 16.3 per cent in other urban areas and 2.7 per cent in rural areas [67].

Research ICT Africa shows why aggregate connection rates require cautious interpretation. Its work identifies persistent inequalities in affordability, device ownership, quality of connection, intensity of use and digital skills [68]. These differences affect the practical ability of households and institutions to use data-intensive services even when nominal internet access is available.

The evidence supports a distinction between connection and effective participation. It does not allow individual AI use to be inferred from household statistics, nor does it show how connectivity conditions affect particular public services or organizations. A fuller analysis would require sectoral and organizational evidence on bandwidth, device access, service affordability, reliability and the support available to users in schools, clinics, municipal offices and other institutions.

5.2.4. Policy verification and institutional routines

South Africa's National AI Policy Framework, released for comment in October 2024, connected AI policy to economic growth, social development, skills, research, regulation and inclusion [69]. A draft National AI Policy was subsequently published in the Government Gazette in April 2026 [70].

The government withdrew the draft after fictitious references were identified in its bibliography. Reuters reported that the minister described the incident as a failure affecting the credibility and integrity of the policy process [71]. This episode provides direct evidence of a breakdown in source verification within one policy-development process.

The withdrawal does not establish the overall capacity of the South African state, nor does it justify conclusions about every institution involved in AI governance. It does show that credible policy development depends on routines for checking evidence, documenting sources, disclosing the use of generative AI and coordinating expert review before a document reaches formal publication.

Further analysis would need to establish whether such routines exist elsewhere in government, how responsibility for verification is distributed and whether departments possess the staff, time and expertise required to evaluate technical evidence. The episode therefore identifies a specific institutional process for investigation rather than a general diagnosis of state capacity.

5.2.5. What remains to be established

The available evidence documents South Africa's position as a regional location for cloud infrastructure, the existence of an established AI research community, persistent inequalities in connectivity, several infrastructure-related governance questions and a failure of verification in one national policy process [8-10, 62-71].

It provides a less complete account of how these elements connect. The reviewed sources do not establish:

- whether universities, smaller firms and public bodies obtain affordable and sustained access to locally hosted compute;

- how public cloud contracts distribute control, audit rights, portability and long-term dependence;
- whether university-based expertise is routinely incorporated into government programmes and organizational careers;
- how municipalities assess the resource requirements and public benefits of data-centre developments;
- how connectivity constraints affect AI-related service use in particular institutions and communities;
- whether evidence-verification and AI-disclosure procedures are embedded across policy-development processes.

Answering these questions would require administrative and procurement data, institutional case studies, interviews with public agencies and infrastructure providers, municipal planning records and sector-specific research on access and use. Until such evidence is available, the South African case supports conclusions about infrastructure presence, research assets and several conditions that may affect capacity formation. It does not support a comprehensive assessment of how far these assets have been converted into broadly usable and institutionally governed capacity.

5.3. Comparing the evidentiary profiles

India and South Africa differ not only in their institutional configurations but also in the kinds of evidence that are publicly visible.

India's information environment provides detailed administrative statistics on national programmes, digital transactions, public platforms and resources mobilized through state initiatives [15, 16, 54–60]. These sources make programme scale highly visible. They provide a more limited view of allocation, regional distribution, organizational use, contractual control and long-term institutional maintenance. Independent reporting can challenge or qualify official claims, as the evidence on GPU allocation and utilization demonstrates [59], but such verification remains uneven.

South Africa presents a more fragmented evidentiary profile. The available material includes corporate announcements, official household statistics, academic research, policy documents and reporting on infrastructure and policy disputes [8-10, 62-71]. This creates a more heterogeneous picture of the institutions and conflicts involved, but it does not offer comparable national metrics on compute allocation, programme uptake or sustained organizational use.

The difference in source density should not be mistaken for a difference in capacity. A country with extensive government reporting may appear easier to assess because programmes and inputs are highly visible. A country with fewer centralized indicators may appear less developed even when substantial capabilities exist in firms, universities or local institutions. In both cases, the weakest evidence concerns the conversion from formal provision or territorial presence into repeated use, institutional control and durable organizational capability.

The comparison therefore identifies two distinct evidentiary problems. In India, official visibility is high but independent evidence on distribution and use is limited. In South Africa, the evidence is more varied but dispersed across institutions, sectors and individual events. Neither profile provides a sufficient basis for assigning an aggregate level of AI capacity.

5.4. Evidence required for a full diagnosis

A full diagnosis would require evidence that follows resources beyond announcements, programme descriptions and infrastructure presence. It would need to establish who receives access, under what conditions, how resources are used, which institutions retain control and whether temporary provision becomes part of routine organizational practice. Table 5.4 sets out the main evidentiary requirements suggested by the two country applications.

Table 5.4. Evidence required for a fuller diagnosis of AI capacity

Analytical dimension	Evidence currently available	Evidence still needed	Appropriate research method
Digital public infrastructure	Platform descriptions, transaction volumes, user numbers and formal interoperability arrangements [54-57]	Institutional use of AI built on these systems; control over models, cloud services and downstream applications; user redress	Administrative-data analysis; procurement analysis; institutional case studies; user research
Compute access	Announced, onboarded or territorially hosted capacity [8–10, 16, 58]	Allocation times, prices, utilization rates, user composition, regional distribution and technical support	Administrative data; provider data; interviews with users and operators; organizational case studies
Institutional use	Programme descriptions, strategies, pilot announcements and selected implementation reports [16, 58, 69, 70]	Routine use, recurring budgets, organizational ownership, staff responsibilities and continuity after initial funding	Longitudinal organizational case studies; budget analysis; interviews; process tracing
Control and dependence	Public strategies, corporate announcements and general procurement rules [8-10, 16, 54, 69, 70]	Contracts, licences, audit rights, portability provisions, exit conditions and data-governance arrangements	Procurement and contract analysis; interviews with purchasing bodies; regulatory analysis
Language capacity	Public platforms, datasets, benchmarks and openly released models [15, 60]	Integration into services, performance across sectors and languages, maintenance arrangements, user experience and institutional ownership	Sectoral case studies; technical evaluation; organizational interviews; user research
Infrastructure and resource governance	Investment announcements, planning disputes, energy projects and local claims [61, 65, 66]	Verified water and energy demand, permitting conditions, cumulative impacts, public-benefit commitments and enforcement	Planning-document analysis; environmental assessment; municipal case studies; community interviews
Connectivity and participation	Household access statistics and research on affordability, devices and skills [67, 68]	Institutional connectivity, service reliability, costs and actual use in schools, clinics, municipalities and smaller firms	Household and organizational surveys; sectoral studies; ethnographic research
Research and skills	Bibliometric evidence, training programmes and university research output [16, 64]	Career pathways, staff retention, access to compute and data, movement between universities and government, institutional demand	Labour-market analysis; institutional mapping; interviews; career-path studies
Policy and regulatory capability	Published strategies, draft policies and documented	Internal verification routines, staffing, expert-review	Process tracing; document analysis; interviews with

	policy processes [31, 32, 69–71]	procedures, disclosure practices and coordination across agencies	officials and external reviewers
Accountability and public participation	Formal policy commitments, reported disputes and public demands for information [49, 61, 66, 69]	Access to explanations, correction and appeal mechanisms, participation in infrastructure decisions and enforcement outcomes	Legal and procedural analysis; case studies; interviews with affected groups

The country applications do not provide a final diagnosis of AI capacity in India or South Africa. Their value lies in showing how the framework changes the interpretation of familiar indicators. Transaction volumes, programme budgets, GPU counts, cloud regions, research publications and policy documents establish that resources and institutional activity exist. They do not establish how those resources are distributed, whether they are repeatedly used, who controls the systems involved or whether organizations can sustain the capabilities created around them.

This distinction gives the framework a methodological function. It identifies the point at which available evidence no longer supports a claim and specifies the data needed to proceed. Section 5 therefore serves as a bridge between the conceptual framework and the research agenda that follows. Instead of assigning countries a capacity score, it converts broad diagnostic categories into empirically answerable questions about access, use, control, institutionalization and accountability.

6. Research agenda for the Global South: From AI governance to capacity formation

The preceding analysis points to a research agenda that begins before formal regulation and extends beyond the adoption of AI applications. National strategies, data-centre investments, public-sector pilots and growing numbers of users indicate that AI is spreading. They reveal much less about the ability of countries to influence the infrastructures, institutions and contractual relationships through which this diffusion occurs. The central research problem is therefore how AI activity is converted (or fails to be converted) into durable capacity.

This shift does not displace established work on ethics, safety and regulation. It locates those concerns within the political economy through which AI becomes embedded in particular societies. The proposed agenda is organized around six questions: the route through which AI first becomes consequential; the capacities generated by that route; the connections and institutional breaks that follow; the places and actors through which capacity is governed; the distribution of bargaining power; and the relationship between AI investment and locally defined development priorities.

The agenda is particularly relevant to the Global South because incorporation into the AI economy takes place under marked asymmetries. Advanced chips, hyperscale cloud infrastructure, frontier models and investment are concentrated among a small number of firms and countries [31, 32, 41]. Yet this concentration does not eliminate agency. Governments, universities, firms, municipalities, regional organizations and language communities shape how infrastructure is sited, how systems are procured, which data are made available and which public needs receive attention. The empirical task is to identify the forms of capacity and

dependency that emerge from these negotiations rather than assume either technological autonomy or passive dependence in advance.

6.1. Entry point: through what route does AI become consequential?

The first question reconstructs the route through which AI acquires economic, institutional or infrastructural significance. The entry point may lie in commercial adoption, cloud and data-centre investment, public administration, an industrial sector, language technology or a development programme. These routes are analytical starting points rather than permanent country classifications. Their value lies in showing which institutions and material systems shape the trajectory before AI becomes a distinct field of policy.

An entry-point analysis begins with sequence and attachment. It asks which existing systems AI builds upon, which actors make the earliest consequential decisions and what dependencies are introduced at that stage. Cloud contracts, data-centre permits, identity and payment systems, sectoral automation, language-corpus projects and donor-funded pilots may all establish durable conditions before a national AI strategy is written. Starting the inquiry with regulation alone would therefore begin too late.

The research should establish:

- where AI first changes investment, administrative practice or infrastructure;
- which pre-existing institutions and resource systems it uses;
- who controls the initial models, data, contracts and technical standards;
- which capabilities and dependencies are created at this stage; and
- how the initial route shapes subsequent policy choices.

The entry-point question supplies a genealogy of AI capacity formation. It replaces the assumption of a uniform transition with an account of how different political economies encounter AI through different institutional doors.

6.2. Capacity formation: what does diffusion leave behind?

The second question concerns conversion. It asks whether AI diffusion leaves behind resources and institutions that local actors can continue to use, adapt and govern. A country may experience rapid adoption without developing evaluation expertise, host data centres without broadening access to compute, or train specialists without creating organizations able to retain them. Activity and capacity must therefore be observed separately.

Capacity formation is best treated as a process distributed across domains. Computational access, language resources, public procurement expertise, evaluation procedures and sector-specific research may develop at different rates and in different institutions. Some remain enclosed within a company, a city or a donor-funded project. Others become shared resources that strengthen universities, public agencies and local firms. The analytical issue is not simply the quantity of resources but their durability, accessibility and capacity to travel across institutional boundaries.

This question directs empirical attention to the mechanisms of conversion. Researchers should follow the path from infrastructure to access, from training to employment and research careers, from datasets to usable systems, and from experiments to ordinary budgets and organizational routines. They should also identify the conditions under which conversion stops. The five diagnostic gaps developed in Section 4 provide a starting framework: adoption without

control, infrastructure without access, data without usability, skills without institutions, and pilots without systems.

The relevant evidence includes who can use a resource, how it is financed, whether it can survive a change of vendor or donor, which organization assumes responsibility and whether the knowledge required for maintenance remains locally available. Capacity is durable when it can be exercised repeatedly rather than recreated for each project.

6.3. Connection: how are capacities assembled across domains?

The third question examines the institutional and material relationships through which separate capabilities become a system. The policy-capacity literature is useful here because it treats effective action as a combination of analytical, operational and political competences distributed across individual, organizational and system levels [66]. AI capacity poses a related coordination problem, but one that also crosses infrastructures and resource regimes.

Fragmentation can arise even where every relevant component is present. Digital ministries may draft AI strategies while energy and water authorities decide the conditions for data-centre expansion. Universities may produce language datasets that public-service platforms do not use. Public agencies may commission pilots without procurement routes for maintenance. Research groups may have technical expertise but no reliable access to compute. The failure is located at the junction between institutions rather than within any one organization.

Connection should therefore be studied through concrete interfaces: compute allocation between providers and researchers; data-sharing arrangements between public agencies; coordination between infrastructure policy and resource planning; movement of specialists between universities, firms and the state; and the incorporation of successful prototypes into procurement and budget systems. These interfaces reveal who performs the work of coordination, which mandates overlap and where responsibility disappears.

Comparative analysis can then focus on patterns rather than scores. Countries with similar levels of investment may connect resources in very different ways. India's problem centres on linking scale, digital public infrastructure, shared compute and linguistic diversity. South Africa's centres on connecting regional cloud infrastructure to resource governance, public-sector capability and uneven access. Other configurations will emerge where AI enters through manufacturing, agricultural systems, extractive industries or donor-supported public services.

6.4. Place and agency: where is AI capacity governed?

The fourth question moves below and beyond the national state. National strategies and laws remain important, but AI capacity is assembled across cities, provinces, universities, utilities, technology parks, research laboratories, firms, civic networks and infrastructure sites. These actors govern essential conditions even when they do not describe their decisions as AI policy.

The geography is especially visible around data centres. Land-use decisions, grid connections, water supply, environmental assessment, fibre access and backup generation are often negotiated through municipal and provincial institutions. A city that approves an infrastructure project or conditions its access to scarce resources is participating in AI governance. Universities perform a different function by training specialists, maintaining

datasets, evaluating systems and linking international research networks to local problems. Firms and open-source communities may control tools and knowledge that formal policy bodies lack. Civil-society organizations and affected communities can expose resource pressures, discriminatory effects or procedural failures that remain invisible at the national level.

Research should map where authority, expertise and infrastructure are concentrated. It should examine whether metropolitan clusters diffuse capability to other regions or intensify spatial inequality. The object of study is not the city as a substitute for the state, but the distribution of governing functions across levels and organizations.

This dimension can be investigated through three linked questions. Which places function as infrastructure, research or application nodes? Which actors exercise authority over the resources and standards on which those nodes depend? What arrangements allow local concerns and knowledge to influence national and corporate decisions? The result is a more accurate geography of AI capacity than national indicators alone can provide.

6.5. Bargaining power: who sets the terms?

The fifth question asks who determines the conditions under which models, cloud services, data, infrastructure and expertise are accessed. Bargaining power is not treated here as an abstract claim to sovereignty. It consists of specific institutional capabilities that allow public bodies and other local actors to alter prices, contractual obligations, technical standards, access conditions and the distribution of benefits.

Public procurement is one such mechanism. Because governments usually buy rather than build many of the systems they use, technical specifications and contracts become sites of governance. Audit rights, documentation, portability, liability, human oversight, maintenance arrangements and exit provisions affect the ability of public agencies to retain control after deployment. Empirical research on municipal AI procurement shows how inherited purchasing procedures can obscure responsibility and leave public bodies unable to anticipate or govern algorithmic harms [42]. Procurement can also serve as an innovation-policy instrument when public demand is used deliberately to shape markets and technological trajectories [67].

Infrastructure and resource decisions create another source of leverage. Data-centre developers require land, electricity, water, permits and often public incentives. These dependencies give host governments and municipalities an opportunity to negotiate disclosure, grid investment, skills programmes, research access or locally available compute. Such conditions require administrative and technical expertise; without it, formal authority over permits may produce little effective bargaining power.

Data governance, standards and finance create further mechanisms. Stewardship and licensing arrangements determine whether public and language data can be enclosed or used for wider public value. Interoperability and portability requirements constrain switching costs. Public investment and development finance can support shared compute, research networks and open language resources rather than subsidize adoption alone. At the regional level, pooled procurement, common standards and shared infrastructure may enlarge the negotiating unit available to individual countries. The AU Data Policy Framework, for example, explicitly seeks harmonized governance and a shared African data space [68], while the Continental AI Strategy frames coordination and capacity-building as parts of an Africa-centred development agenda [69].

Table 6.1 operationalizes these mechanisms. It is not intended as another national index. Its purpose is to identify where negotiating capability is located, which instruments exist and whether they can be exercised in practice.

Table 6.1 Mechanisms operationalisation

Mechanism	Leverage	Illustrative instruments	Evidence of effective use
Procurement	Public purchasing power	Audit and documentation clauses; portability; liability; exit rights; local maintenance	Technical requirements appear in tenders; agencies can inspect, challenge and replace systems
Compute	Allocation of scarce computational resources	Public compute; research credits; university consortia; regional pools; access conditions in investment agreements	Universities, start-ups and public bodies receive affordable, sustained access
Data governance	Control over access, reuse and public value	Stewardship; licensing; sectoral data spaces; community governance; cross-border rules	Access conditions are documented, enforceable and linked to public or community benefit
Infrastructure and resources	Authority over land, electricity, water, permits and incentives	Disclosure; grid and water conditions; community benefits; research or skills commitments	Approvals specify measurable public obligations and authorities can monitor compliance
Finance and investment	Ability to condition public support and external capital	Incentive design; co-investment; technology transfer; research partnerships; development finance	Investment leaves durable skills, institutions, infrastructure or locally controlled knowledge
Standards and interoperability	Reduction of lock-in and switching costs	Open APIs; portability; common documentation and evaluation rules	Systems can exchange data, be independently evaluated and migrate between providers
Regional cooperation	Pooling of demand, infrastructure and regulatory influence	Joint procurement; shared compute; common standards; regional audit or research centres	Cooperation produces operational facilities, contracts or enforceable rules rather than declarations alone

Bargaining mechanisms are most consequential when they reinforce one another. Procurement conditions are stronger when agencies possess technical expertise and viable alternatives. Data-centre conditions are more credible when energy and planning authorities coordinate with digital-policy bodies. Regional standards matter when they are incorporated into tenders and investment agreements. The research task is therefore to trace not only the existence of an instrument but the institutional coalition capable of using it.

6.6. Development alignment: what purposes and whose interests shape capacity?

The sixth question asks how AI capacity relates to locally defined development priorities. The relevant benchmark is not whether a country reproduces the technological profile of a

frontier-model producer. It is whether the capabilities being built support public services, sectoral upgrading, local-language access, research, employment, climate adaptation or other priorities established through legitimate political processes. UNCTAD's emphasis on infrastructure, data and skills as leverage points for inclusive development supports this wider view of AI foundations [19]. The AU Continental AI Strategy similarly places development and prosperity at the centre of its regional agenda [69].

Developmental purpose does not, however, guarantee accountable outcomes. AI introduced through social protection, health, education, taxation, migration or urban management can strengthen administrative capability while also expanding classification, surveillance and centralized control. The UNESCO Recommendation on the Ethics of Artificial Intelligence places human rights, transparency, accountability, human oversight and the possibility of redress among the safeguards required across AI systems [70]. These concerns belong inside the analysis of capacity formation rather than being added after deployment.

Development alignment should therefore be evaluated along two axes. The first concerns direction: which sectors, populations and institutional capabilities receive sustained investment? The second concerns governance: can affected people know that AI is being used, correct relevant data, challenge decisions and obtain human review? A system may address a genuine public need while remaining procedurally opaque. Conversely, a well-governed pilot may have little developmental significance if it cannot be sustained or reaches only a narrow population.

The distributive analysis should extend to resource costs and spatial concentration. Researchers should ask where benefits accrue, which institutions bear maintenance costs, whose data and languages are represented, and whether infrastructure investment widens or narrows regional disparities. Development alignment is present when AI capability is connected to public purpose, institutional accountability and a credible pathway for distributing benefits.

Table 6.2 identifies recurring risks in state-led and development-oriented AI. These risks do not constitute an argument against public-sector capacity. They specify the safeguards without which capacity can become administrative power that is difficult to contest.

Table 6.2 Recurring risks in state-led and development-oriented AI

Risk	Institutional expression	Research indicators
Administrative opacity	AI-assisted decisions cannot be understood or challenged	No notice, explanation, public register, audit record or accessible appeal
Surveillance expansion	Service systems enable broader monitoring or secondary uses	Cross-sector linkage, weak purpose limitation, continuous tracking or unclear retention
Concentration of authority	Integrated data and AI systems centralize discretion	Weak external oversight, limited local discretion and unclear institutional checks
Vendor-mediated governance	Public judgment depends on proprietary systems and expertise	Trade-secrecy barriers, absent audit rights, limited documentation and high switching costs
Inequality at scale	Automated systems reproduce unequal access and categories	Disparate outcomes across regions, languages, income groups or service users
Weak contestability	Affected people cannot correct data or obtain human review	No practical redress, correction process, review timetable or responsible institutional owner

6.7. Research design implications

Studying capacity formation requires methods capable of following AI across documents, organizations and material infrastructures. Policy analysis remains necessary but cannot by itself reveal contractual dependence, compute access, data quality, resource allocation or the organizational fate of pilots.

A research design should combine four moves. The first is process tracing: reconstructing when AI became consequential and how decisions in one domain opened or constrained later choices. The second is relational mapping: identifying the organizations that control infrastructure, finance, data, expertise and authorization, together with the formal and informal connections among them. The third is junction analysis: examining procurement, permitting, data-sharing, budgeting and other interfaces where responsibility is transferred or fragmented. The fourth is outcome tracing: following a resource or project far enough to determine whether it becomes a durable capability.

The evidence base may include contracts and tenders, infrastructure and environmental approvals, budget documents, technical standards, data-governance instruments, organizational charts, programme evaluations, interviews and observations of implementation. Researchers should distinguish announced commitments from operational resources and formal mandates from capabilities that can actually be exercised. This is particularly important in fast-moving policy fields, where strategies and investment figures can create an appearance of capacity before institutions and access arrangements exist.

Comparison should be organized around configurations rather than rankings. A structured comparison can use the same six questions across cases while allowing different entry routes, actors and outcomes to emerge. This preserves comparability without treating all countries as incomplete versions of a single technological model.

6.8. Toward a Global South agenda on AI capacity

This agenda does not replace research on AI ethics, safety, innovation or regulation. It changes the unit of analysis. The risks at stake arise from the way the underlying capacity is assembled. Vendor dependence, exclusion from compute, weak data institutions, resource pressure, linguistic invisibility and fragmented implementation shape the range of choices available before a particular system is assessed for safety or fairness.

Table 6.8 Research dimentions

Research dimension	Core question	Analytical contribution
Entry point	Through what route does AI first become consequential?	Identifies the initial actors, infrastructures and dependencies
Capacity formation	What durable capabilities does diffusion leave behind?	Distinguishes adoption, hosting and projects from sustained capacity
Connection	Which resources and institutions are linked, and where do they fragment?	Locates the junctions at which capacity is assembled or blocked
Place and agency	Where is capacity governed, and by which state and non-state actors?	Reveals subnational geographies and distributed authority

Bargaining power	Who sets the terms, and through which instruments can they be changed?	Makes dependency, value capture and negotiating capability observable
Development alignment	Which priorities and populations does capacity serve?	Connects AI investment to public value, distribution and accountability

For countries of the Global South, governing AI therefore includes governing the conditions under which AI becomes part of economic and public life. Those conditions encompass compute, data, language, infrastructure, natural resources, institutions, skills, contracts and regional relationships. The proposed research programme asks where these elements come together, who controls the connections and what forms of public and developmental agency they make possible.

The next frontier of AI governance in the Global South is the study of capacity formation: the processes through which countries are incorporated into the AI economy, the dependencies and capabilities produced through that incorporation, and the institutional means available to reshape its terms.

7. Closing: From AI adoption to the conditions of AI futures

The expansion of AI across the Global South is increasingly visible. Governments announce national programmes, firms build cloud regions and data centres, public agencies adopt automated systems, and universities develop models and language resources. These developments are often treated as evidence that countries are becoming more capable in AI. The analysis in this paper suggests a more demanding conclusion: visible activity shows that AI is spreading, but it does not establish who can shape the systems through which that spread occurs.

This distinction changes the object of analysis. The central question is not simply whether a country has AI applications, infrastructure or policy commitments. It is whether institutions can influence access, technical design, contractual terms, resource allocation and accountability.

The framework developed here therefore treats AI capacity as a problem of conversion and connection. Investment must be converted into usable infrastructure; infrastructure into access; data and models into organizational practice; expertise into institutional authority; and experimentation into systems that can be maintained, evaluated and contested. Failure can occur at any of these points. The five diagnostic gaps identify these failures, but their deeper significance lies in showing that capacity cannot be inferred from inputs alone.

This also reveals a governance problem that national AI strategies often obscure. The decisions shaping AI are dispersed across institutions that may not carry an explicit AI mandate. Procurement bodies determine contractual dependence. Energy regulators and municipalities shape the material conditions of data-centre expansion. Universities and research institutes influence access to expertise. Cloud providers set technical and commercial conditions. Public agencies decide whether systems become embedded in routine administration. AI governance therefore depends on the ability to coordinate institutions whose mandates, resources and accountability structures were not designed around AI.

The implications for policy are concrete. Governments need evidence that follows programmes beyond announcement and provision. Compute initiatives should be assessed through allocation, price, utilization and user composition. Infrastructure investment should be examined through local access, resource commitments and public-benefit conditions. Language

programmes require evidence of service integration, performance and maintenance. Public-sector adoption should be evaluated through contracts, audit rights, portability, staffing and long-term organizational ownership. Without this evidence, policy remains oriented toward mobilizing inputs while lacking a reliable account of what those inputs produce.

The broader implication is that the position of countries in the AI economy will be determined less by whether they adopt AI than by the terms under which adoption takes place. Those terms concern ownership, access, standards, contracts, resource use, institutional authority and the ability to contest decisions. Countries that cannot negotiate these conditions may experience rapid diffusion while remaining dependent on external infrastructures, suppliers and models. Countries that build capacity to connect resources, institutions and public authority gain greater scope to determine where AI is used, whose interests it serves and which risks are accepted.

AI capacity should therefore be understood as a form of collective institutional agency. Its purpose is not technological self-sufficiency in every component of the AI stack. It is the ability to make informed choices, negotiate dependencies, coordinate fragmented authority and preserve meaningful control over systems that increasingly shape economic and social life. This is the threshold that separates participation in AI diffusion from the capacity to shape an AI future.

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